# Week 1

## DevOps Internship

**Release Date: 27 July 2026**

This week, you'll begin with **Linux Fundamentals, Linux Administration & Networking Basics**. Mastering Linux is the first step toward managing servers, automating workflows, and building modern DevOps infrastructure.

**Topic:- Linux Fundamentals, Linux Administration & Networking Basics**

This week you are expected to learn:

- Introduction to Linux & Open Source
- Linux File System Hierarchy (FHS)
- Ubuntu Installation (VirtualBox/VMware/WSL)
- File & Directory Management
- File Permissions & Ownership
- Users & Groups Management
- Linux Package Management
- Process Management
- Environment Variables
- Basic Networking Concepts
- SSH Fundamentals
- Common Linux Commands

***Learning Approach**

Focus on building a strong understanding of Linux fundamentals rather than simply completing the tasks. Spend time practicing commands, exploring the Linux environment, and understanding why each concept is important. Hands-on practice, curiosity, and consistency will help you gain confidence and prepare you for the advanced DevOps tools covered in the upcoming weeks.

***Keep these points in mind:**

- ✓ Practice every command instead of just reading about it.
- ✓ Understand the purpose of each concept before moving forward.
- ✓ Maintain notes for future reference.
- ✓ Verify AI-generated answers through practical implementation.
- ✓ Complete all activities before the submission deadline.

## Weekly Task

***Complete the following practical exercises:**

- Install Ubuntu using VirtualBox, VMware, or WSL.
- Explore the Linux File System.
- Create and manage users and groups.
- Practice file and directory management.
- Configure file permissions and ownership.
- Install and update packages using the package manager.
- Monitor running processes.
- Configure environment variables.
- Practice networking commands such as:
  - hostname
  - hostnamectl
  - ip addr
  - ping
  - ifconfig
- Connect to a system using SSH (if available).
- Prepare a short PDF explaining:
  - Linux Fundamentals
  - Linux File System
  - Users & Groups
  - File Permissions
  - Package Management
  - Environment Variables
  - Basic Networking
  - SSH
  - 20 Common Linux Commands

**Task Submission**

Once you complete the above task, submit:

- ✓ PDF Report (2–4 pages)
- ✓ Screenshot of Ubuntu Installation
- ✓ Screenshot of Linux Terminal
- ✓ Screenshot of Users & Groups
- ✓ Screenshot of File Permissions
- ✓ Screenshot of Networking Commands

**Task Submission Deadline: Wednesday (EOD)**

**Scenario**

Imagine you have joined a company as a **Junior DevOps Engineer**. Your manager asks you to prepare a Linux server for a new development team by configuring users, directories, permissions, and basic networking.

***Complete the following:**

- Create a **DevOps** directory.
- Inside it, create:
  - Projects
  - Scripts
  - Logs
  - Backup
- Create sample files inside each folder.
- Create two Linux users and one group named **developers**.
- Add both users to the **developers** group.
- Assign appropriate file permissions.
- Configure your system hostname.
- Verify internet connectivity using the ping command.
- Display your IP address and system information.
- Compress the **DevOps** directory into a ZIP or TAR file.
- Document each step performed.

**Activity Submission**

After completing the hands-on activity, submit:

- ✓ ZIP/TAR file of the DevOps directory
- ✓ Screenshot of Directory Structure
- ✓ Screenshot of Users & Groups
- ✓ Screenshot of File Permissions
- ✓ Screenshot of Network Connectivity
- ✓ Short implementation summary (4–5 lines)

**Activity Submission Deadline: Friday (EOD)**

 

## AI Assistant Task

Use an AI Assistant such as **ChatGPT, GitHub Copilot, Amazon Q, Gemini, Claude, or Microsoft Copilot** to strengthen your understanding of Linux.

***Complete the following:**

- Ask AI to explain Linux administration in simple terms.
- Generate a cheat sheet for **20 commonly used Linux commands**.
- Ask AI to explain Linux users, groups, and file permissions.
- Ask AI to explain SSH and its role in DevOps.
- Generate a Linux troubleshooting scenario and try solving it.
- Verify the AI-generated commands by executing them in your Linux environment.

Take time to understand what each Linux command does rather than simply executing it. A strong Linux foundation will make learning Docker, Kubernetes, Jenkins, Terraform, and Cloud technologies much easier.

**If you face any genuine issues, feel free to reach out before the submission deadline.**

 